

DETECTING THE IMPOSTER

How Architecture, Pretraining Scale, and Fine-Tuning Strategy Affect AI-Generated Art Detection

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Motivation

- AI-generated artwork has rapidly improved, making it increasingly difficult to distinguish from human-created art, reliable detection becomes critical for authenticity verification and copyright protection.
- We lack a large-scale, human vs AI dataset.
- Large pretrained foundation models encode rich semantic representations learned from web-scale data, which may help with detecting the difference between human and AI art.
- It remains unclear whether detection performance depends more on: **model architecture**, **pretraining data scale**, or **adaptation strategy**.
- Research Question: How do model architecture (CNN vs. Transformer), pre-training data (ImageNet-1k vs. 21k vs. OpenAI CLIP), and adaptation strategy (Linear Probe vs. Full Fine-Tuning) interact to determine performance when detecting AI-generated art with limited labeled examples?

Models

We compare two widely used visual models:

- ResNet-50 (RN50)
 - Convolutional Neural Network (CNN) architecture
 - Emphasizes local texture and spatial feature extraction
 - ~100MB model size
 - Strong inductive bias toward locality
- Vision Transformer (ViT-B/16)
 - Transformer-based image encoder
 - Divides image into 16×16 patches
 - Models global relationships via self-attention
 - ~330MB model size
 - Strong global context modeling ability

Datasets

- Downstream Task Dataset: Kaggle AI vs Human Art Dataset
 - Original training set size: 950 images
 - Augmented to 6,875 images via randomized cropping, cuttingout, and color filtering
 - Images resized and normalized to the same resolution
 - Held-out test images: 198
 - Balanced class distribution (AI / Human)

- Pretraining Datasets with different scales:

ImageNet-1K	ImageNet-21K	OpenAI CLIP
~1.2M labeled images 1,000 object categories - Standard supervised pretraining baseline	~14M labeled images 21,000 object categories - Larger supervised pretraining scale	~400M image-text pairs - Web-scale multimodal pretraining - Rich semantic alignment between vision and language

Experimental Setup

- The **test set** is held out and used only once for final evaluation. A **validation split** (~85/15) is created from the augmented training data.
- We apply standard ImageNet preprocessing consisting of **resizing**, **center cropping** to 224×224, **conversion** to tensor, and **normalization** with channel-wise mean and std.
- Models are initialized from three pretraining sources: **ImageNet-1K**, **ImageNet-21K**, and **OpenAI CLIP** (400M). In all cases, the original classification head is replaced with a **binary linear layer** mapping features to two classes (AI vs. Human).
- For each configuration, we compare two training modes. In **Linear Probe**, the base model is frozen and only the classifier is trained. In **Full Fine-Tuning**, all parameters are updated end-to-end.
- Models are trained using **cross-entropy** loss and the **Adam** optimizer for **10 epochs**. The best checkpoint of a model is selected based on **validation F1 score**. We use default values (specified by timm and open_clip libraries) for hyperparameters, including learning rate and momentum.
- Performance is evaluated using **Accuracy**, Precision, Recall, **F1 score**, and Confusion Matrix on the held-out test set.

Results

Table 1: Validation and Test Performance Across All Configurations

Model	Pretraining	Mode	Val Acc	Val F1	Test Acc	Test F1
RN50	ImageNet-1K	Linear Probe	0.8623	0.8704	0.7653	0.7629
RN50	ImageNet-1K	Full Fine-Tune	0.9884	0.9889	0.8163	0.7907
RN50	ImageNet-21K	Linear Probe	0.9748	0.9738	0.7755	0.7826
RN50	ImageNet-21K	Full Fine-Tune	0.9893	0.9890	0.8061	0.7865
RN50	OpenAI CLIP	Linear Probe	0.8623	0.8704	0.7857	0.7789
RN50	OpenAI CLIP	Full Fine-Tune	0.9884	0.9889	0.8265	0.8172
ViT-B/16	ImageNet-1K	Linear Probe	0.8865	0.8926	0.7857	0.7470
ViT-B/16	ImageNet-1K	Full Fine-Tune	0.9893	0.9897	0.8367	0.8182
ViT-B/16	ImageNet-21K	Linear Probe	0.9224	0.9226	0.7959	0.7826
ViT-B/16	ImageNet-21K	Full Fine-Tune	0.9903	0.9900	0.8061	0.7865
ViT-B/16	OpenAI CLIP	Linear Probe	0.8865	0.8926	0.8980	0.8889
ViT-B/16	OpenAI CLIP	Full Fine-Tune	0.9893	0.9897	0.8367	0.8367

Table 2: Average Training Time per Mode

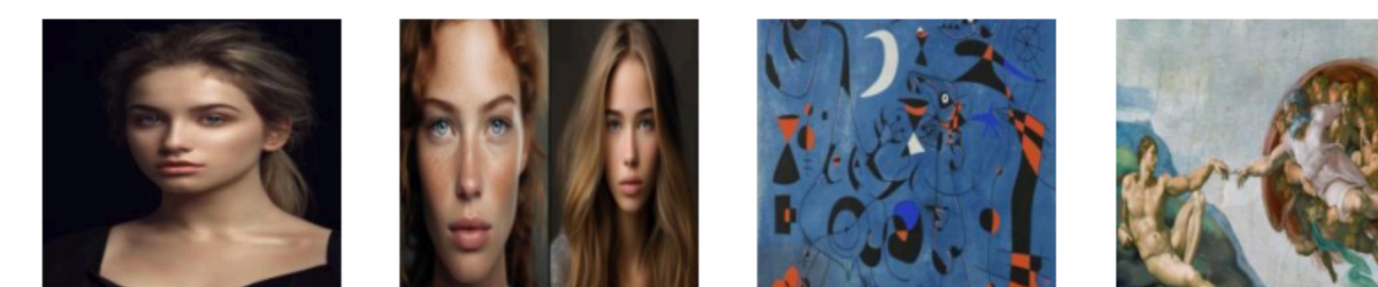
Training Mode	Avg Train Time (sec)
Linear Probe	118
Full Fine-Tune	125

Discussion

- ViT-B/16 consistently outperforms RN50** on the test set across pretraining sources. The best overall configuration (ViT-B/16 + OpenAI CLIP + Linear Probe) achieves the highest test F1 score (0.8889), suggesting that Transformer-based architectures better capture global structural patterns in AI-generated art compared to CNNs, which emphasize local textures.
- Larger pretraining sources** (ImageNet-21K and OpenAI CLIP 400M) **improve validation performance** across both backbones. OpenAI CLIP pretraining yields the strongest test generalization. This indicates that web-scale multimodal training provides more transferable representations for artistic domains.
- Full fine-tuning achieves near-perfect validation F1** (~0.99) across configurations, but this does not consistently translate to superior test performance. In contrast, **Linear Probe achieves the best overall test result**. This suggests that full fine-tuning may overfit to dataset-specific features, while frozen pretrained representations preserve stronger generalization.
- Linear Probe also offers lower computational cost**. Average training time per configuration is shorter (~118s) compared to Full Fine-Tuning (~125s), while achieving competitive or superior test performance in the best case. This highlights a favorable trade-off between efficiency and generalization when strong pretrained features are available.
- The observed validation-test discrepancy, especially for fully fine-tuned models, indicates **sensitivity to distribution shift**. Aggressive adaptation may reduce robustness on unseen samples.

Future Work

- How does larger-scale multimodal pretraining influence robustness? Would models pretrained on web-scale datasets (e.g., LAION) further improve generalization in AI-art detection compared to ImageNet or CLIP 400M?
- How robust are detection models under distribution shift? How do performance and generalization change across unseen artistic styles, new generative models, resolution changes, or image perturbations?
- Can we balance adaptability and efficiency more effectively? Do parameter-efficient fine-tuning methods (e.g., LoRA) provide the adaptability of full fine-tuning while preserving the generalization benefits observed in linear probing?



References

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